Highlights

Porous polyimide/hexagonal boron nitride films by non-solvent induced phase separation: a three-phase Lewis–Nielsen description of dielectric and thermal properties

- Porous PI/h-BN films made by NIPS over the full 0–70 wt% h-BN range
- Three-phase Lewis–Nielsen model unifies dielectric and thermal behavior
- Pore shape factor $A = 0.08$ fingerprints NIPS sponge vs. spherical pores
- Single-parameter thermal fit: $\lambda_{\mathrm{BN,eff}} = 6.8\,\mathrm{W\,m^{-1}\,K^{-1}}$, interface-limited
- 70 wt% h-BN gives the best measured ε_r–λ balance; model loci agree